DES PUNE UNIVERSITY
School of Engineering and Technology
Computer Engineering and Technology
Program: B.Tech. Computer Science and Engineering

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Subject: Java		
Assignment No.: 3	Title: Employee Management System	
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Java
Assignment - 3

- Aim:

Implement a class hierarchy using all types of inheritance and demonstrate method overriding in the context of an employee management system.

- Theory:

Inheritance is an OOP feature in Java where one class (subclass) can acquire the properties and methods of another class (superclass). It is implemented using the 'extends' keyword (for classes) and 'implements' keyword (for interfaces)

1. Single inheritance

One child class inherits from one parent class.

Parent
↓
Child

Syntax:

```
class Parent
{
    void display()
    {
        // statement
    }
}
class child extends Parent {
    void display () { // statement }
}
```

```
    void fn () {     }
}
class Grandchild extends Child {
    void fn () {     }
}
```

4. Multiple inheritance
A class implements multiple interfaces at the same time. It is not supported with classes to avoid ambiguity but possible through interface.

Interface 1   Interface 2

Child

5. Hybrid inheritance
Combination of two or more types of inheritance. It is a mix of extends and implements.

Method Overriding
Method overriding occurs when a child class provides its own implementation of a method that is already in its parent class. @override annotation is commonly used to indicate overriding. It only works in an IS-A relationship.
Syntax:

```
class Parent {
    void show() {   }
}
class Child extends Parent {
@override
void show () {   }
}
```

2. Hierarchical Inheritance
Multiple child classes inherits from the same parent class.

Parent
↙ ↘
Child 1 Child 2

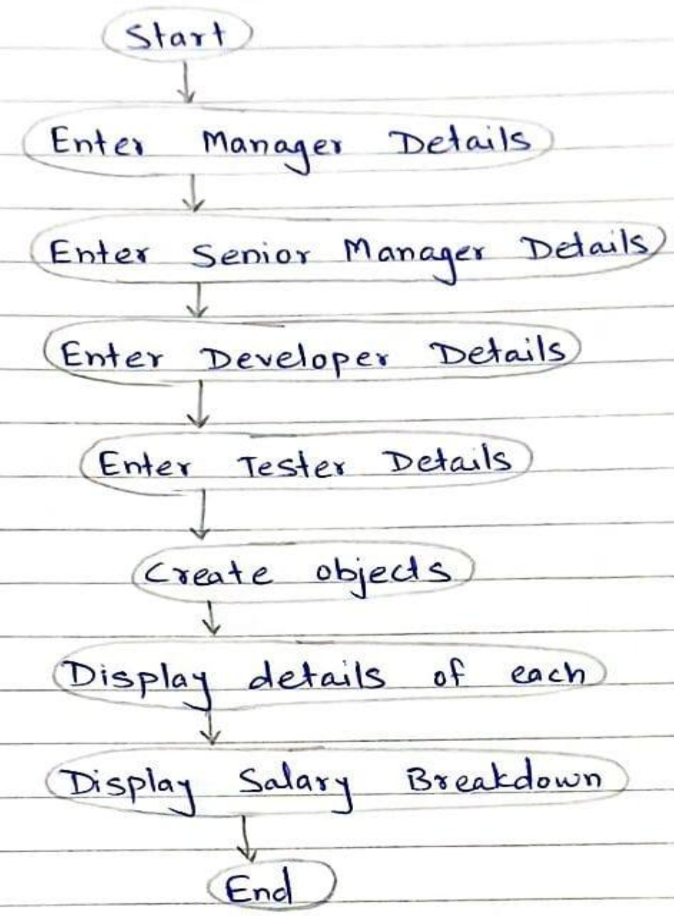
Syntax:
```
class Parent {
    void fn () { //statement }
}
class Child1 extends Parent {
    void fn() {//statement}
}
class Child2 extends Parent {
    void fn() { //statement }
}
```

3. Multilevel inheritance
A chain of inheritance where a class inherits from another child class.

Parent
↓
Child
↓
Grandchild

Syntax:
```
class Parent {
    void fn () {  }
}
class Child extends Parent {
```

Flowchart:



Code:

```
package Assignment3;

import java.util.Scanner;

class Employee
{
    int empID;
    double basicSalary;
    String name, department;

    Employee(String name, int empID, double basicSalary, String department)
    {
        this.name=name;
        this.empID=empID;
        this.basicSalary=basicSalary;
        this.department=department;
    }

    void display()
    {
        System.out.println("Name: "+name);
        System.out.println("Employee ID: "+empID);
        System.out.println("Department: "+department);
        System.out.println("Basic Salary: "+basicSalary);
    }
}
```

```java
class Manager extends Employee
{
    double HRA;
    double DA;
    double award;

    Manager(String name, int empID, double basicSalary, String department, boolean outstanding)
    {
        super(name, empID, basicSalary, department);
        this.HRA= 0.20*basicSalary;
        this.DA= 0.10*basicSalary;
        if(outstanding)
        {
            this.award = 5000;
        }
        else
        {
            this.award = 0;
        }
    }

    @Override
    void display()
    {
        super.display();
        System.out.println("HRA: "+HRA);
        System.out.println("DA: "+DA);

        System.out.println("Outstanding Award: "+award);
    }
}

class SeniorManager extends Manager
{
    int teamSize;

    SeniorManager(String name, int empID, double basicSalary, String department, boolean outstanding, int te
    {
        super(name, empID, basicSalary, department, outstanding);
        this.teamSize = teamSize;
    }

    @Override
    void display()
    {
        super.display();
        System.out.println("Team Size: "+teamSize);
    }
}

class Developer extends Employee
{
    String language;
    double HRA;
    double DA;

    double award;

    Developer(String name, int empID, double basicSalary, String department, String language, boolean outsta
    {
        super(name, empID, basicSalary, department);
        this.language= language;
        this.HRA= 0.15*basicSalary;
        this.DA= 0.08*basicSalary;
        if(outstanding)
        {
            this.award = 3000;
        }
        else
        {
            this.award = 0;
        }
    }

    @Override
    void display()
    {
        super.display();
        System.out.println("Language: "+ language);
        System.out.println("HRA: "+ HRA);
        System.out.println("DA: "+ DA);
        System.out.println("Outstanding Award: "+ award);
    }
```

```
class Tester extends Employee
{
    String tool;
    double HRA;
    double DA;
    double award;

    Tester(String name, int empID, double basicSalary, String department, String tool, boolean outstanding)
    {
        super(name, empID, basicSalary, department);
        this.tool= tool;
        this.HRA= 0.12*basicSalary;
        this.DA= 0.09*basicSalary;
        if(outstanding)
        {
            this.award = 2000;
        }
        else
        {
            this.award = 0;
        }
    }

    @Override
    void display()
    {
        super.display();


        System.out.println("Testing Tool: "+ tool);
        System.out.println("HRA: "+HRA);
        System.out.println("DA: "+DA);
        System.out.println("Outstanding Award: "+award);
    }
}

class AdminPanel
{
    double HRF,PF,PT;

    void SalaryBreakdown(Employee emp, double HRA, double DA, double award)
    {
        double grossSalary = emp.basicSalary + HRA + DA + award;
        this.PF = 0.05* emp.basicSalary;
        this.PT = 0.02* emp.basicSalary;
        this.HRF= 1000;
        double netSalary= grossSalary-(PF + PT + HRF);
        System.out.println("~ Salary Breakdown ~");
        System.out.println("Gross Salary: " + grossSalary);
        System.out.println("PF: " + PF);
        System.out.println("PT: " + PT);
        System.out.println("HRF: " + HRF);
        System.out.println("Net Salary: " + netSalary);
    }
}


public class EmployeeManagementSystem
{
    public static void main(String[] args)
    {
        Scanner sc= new Scanner(System.in);
        AdminPanel admin = new AdminPanel();

        System.out.println("Enter Manager Details");
        System.out.print("Name: ");
        String mName = sc.nextLine();
        System.out.print("Employee ID: ");
        int mID=sc.nextInt();
        System.out.print("Basic Salary: ");
        double mSalary = sc.nextDouble();
        sc.nextLine();
        System.out.print("Department: ");
        String mDepartment = sc.nextLine();
        System.out.print("Is Performance Outstanding?(true/false): ");
        boolean mOutstanding = sc.nextBoolean();
        sc.nextLine();
        Manager m = new Manager(mName, mID, mSalary, mDepartment, mOutstanding);

        System.out.println("\nEnter Senior Manager Details");
        System.out.print("Name: ");
        String smName = sc.nextLine();
        System.out.print("Employee ID: ");
        int smID=sc.nextInt();
```

```
        System.out.print("Basic Salary: ");
        double smSalary = sc.nextDouble();
        sc.nextLine();
        System.out.print("Department: ");
        String smDepartment = sc.nextLine();
        System.out.print("Is Performance Outstanding?(true/false): ");
        boolean smOutstanding = sc.nextBoolean();
        System.out.print("Team Size: ");
        int teamSize = sc.nextInt();
        sc.nextLine();
        SeniorManager sm = new SeniorManager(smName, smID, smSalary, smDepartment, smOutstanding, teamSize);

        System.out.println("\nEnter Developer Details");
        System.out.print("Name: ");
        String dName = sc.nextLine();
        System.out.print("Employee ID: ");
        int dID=sc.nextInt();
        System.out.print("Basic Salary: ");
        double dSalary = sc.nextDouble();
        sc.nextLine();
        System.out.print("Department: ");
        String dDepartment = sc.nextLine();
        System.out.print("Programming Language: ");
        String language = sc.nextLine();
        System.out.print("Is Performance Outstanding?(true/false): ");
        boolean dOutstanding = sc.nextBoolean();
        sc.nextLine();


        Developer dev = new Developer(dName, dID, dSalary, dDepartment, language, dOutstanding);

        System.out.println("\nEnter Tester Details");
        System.out.print("Name: ");
        String tName = sc.nextLine();
        System.out.print("Employee ID: ");
        int tID=sc.nextInt();
        System.out.print("Basic Salary: ");
        double tSalary = sc.nextDouble();
        sc.nextLine();
        System.out.print("Department: ");
        String tDepartment = sc.nextLine();
        System.out.print("Testing Tool: ");
        String tool = sc.nextLine();
        System.out.print("Is Performance Outstanding?(true/false): ");
        boolean tOutstanding = sc.nextBoolean();
        sc.nextLine();
        Tester tester = new Tester(tName, tID, tSalary, tDepartment, tool, tOutstanding);

        System.out.println("\n~ Employee Details ~");

        System.out.println("\nManager:");
        m.display();
        admin.SalaryBreakdown(m, m.HRA, m.DA, m.award);

        System.out.println("\nSenior Manager:");
        sm.display();

        admin.SalaryBreakdown(sm, sm.HRA, sm.DA, sm.award);

        System.out.println("\nDeveloper:");
        dev.display();
        admin.SalaryBreakdown(dev, dev.HRA, dev.DA, dev.award);

        System.out.println("\nTester:");
        tester.display();
        admin.SalaryBreakdown(tester, tester.HRA, tester.DA, tester.award);

        sc.close();
    }
}
```

Output:

```
Enter Manager Details
Name: Riddhee
Employee ID: 1
Basic Salary: 100000
Department: Marketing
Is Performance Outstanding?(true/false): true

Enter Senior Manager Details
Name: Nandini
Employee ID: 2
Basic Salary: 120000
Department: Product
Is Performance Outstanding?(true/false): true
Team Size: 5

Enter Developer Details
Name: Shardul
Employee ID: 3
Basic Salary: 120000
Department: Product
Programming Language: Java
Is Performance Outstanding?(true/false): true

Enter Tester Details
Name: Neeraj
Employee ID: 4


Basic Salary: 120000
Department: Product
Testing Tool: C++
Is Performance Outstanding?(true/false): true

~ Employee Details ~

Manager:
Name: Riddhee
Employee ID: 1
Department: Marketing
Basic Salary: 100000.0
HRA: 20000.0
DA: 10000.0
Outstanding Award: 5000.0
~ Salary Breakdown ~
Gross Salary: 135000.0
PF: 5000.0
PT: 2000.0
HRF: 1000.0
Net Salary: 127000.0

Senior Manager:
Name: Nandini
Employee ID: 2
Department: Product
```

```
Basic Salary: 120000.0
HRA: 24000.0
DA: 12000.0
Outstanding Award: 5000.0
Team Size: 5
~ Salary Breakdown ~
Gross Salary: 161000.0
PF: 6000.0
PT: 2400.0
HRF: 1000.0
Net Salary: 151600.0

Developer:
Name: Shardul
Employee ID: 3
Department: Product
Basic Salary: 120000.0
Language: Java
HRA: 18000.0
DA: 9600.0
Outstanding Award: 3000.0
~ Salary Breakdown ~
Gross Salary: 150600.0
PF: 6000.0
PT: 2400.0
HRF: 1000.0


Net Salary: 141200.0

Tester:
Name: Neeraj
Employee ID: 4
Department: Product
Basic Salary: 120000.0
Testing Tool: C++
HRA: 14400.0
DA: 10800.0
Outstanding Award: 2000.0
~ Salary Breakdown ~
Gross Salary: 147200.0
PF: 6000.0
PT: 2400.0
HRF: 1000.0
Net Salary: 137800.0
```

classmate
Date
Page

Conclusion:
This program demonstrates how to use inheritance & method overriding.